

INJECTED MONOSODIUM GLUTAMATE INCREASES HOLOTRYPTOPHAN PYRROLASE ACTIVITY AND PLASMA CORTICOSTERONE LEVELS MORE IN MALE THAN FEMALE RATS

SAMINA BANO, DARAKHSHAN J. HALEEM AND M.A. HALEEM

Department of Biochemistry, University of Karachi
Karachi-75270, Pakistan

ABSTRACT

Monosodium glutamate (MSG) injected intra-peritoneally at doses 4 g/kg increased holotryptophan pyrrolase activity significantly by 61.6% in male rats. The increases were smaller (24%) and non-significant in females. Apotryptophan was decreased by 61% and comparably in the two sexes. Cumulative effect on total enzyme activity was a reduction of 37.5% in female rats. A decrease of 20.6% in male rats was not significant. Plasma levels of corticosterone were increased significantly in male but not in female rats. The sex differences are explained in terms of female hormones being effected by MSG administration and higher circulating MSG-induced corticosterone levels being involved in the enhancement of holotryptophan pyrrolase activity in male rats. Plasma total tryptophan levels were increased due to an inhibition of total tryptophan pyrrolase activity in females but not in males. Plasma free tryptophan concentrations were decreased in both sexes.

INTRODUCTION

Tryptophan pyrrolase (L-tryptophan-2, 3-dioxygenase EC. 1.13.1.12) is the primary enzyme catabolizing tryptophan in mammals and regulation of its activity may influence tryptophan levels in circulation. Tryptophan levels are of importance because tryptophan is least abundant amino acid in pool available for protein synthesis (Murnro, 1968) and may have a role in the regulation of specific enzyme degradation (Cihak & Lamar, 1973). In addition plasma tryptophan concentration can influence brain 5-hydroxytryptamine (5-HT) (Fernstrom & Wurtman, 1971). Control of tryptophan pyrrolase activity may also be important in regulating the levels of tryptophan metabolites in kynurenine pathway for the biosynthesis of nicotinamide coenzymes (Powanda & Wannemacher, 1971). Curzon & Green (1968) have shown that induction of tryptophan pyrrolase by cortisol can reduce brain 5-hydroxytryptamine levels possibly through action of tryptophan metabolites. Interaction between tryptophan metabolites and 5-HT are also seen in behavioural experiments (Lapin, 1972).

MSG injection has been also reported to reduce serum estrogen levels in female rats and testosterone levels in male rats (Chin Hsu et al., 1990).

We have used single injection of MSG (4 g/kg) to determine its effect on hepatic tryptophan pyrrolase activity and plasma tryptophan and corticosterone levels in male and female rats.

MATERIALS AND METHODS

Locally bred male and female rats weighing 200-250 g were caged in a quiet room were fed on cubes of standard rat food for at least five days before starting the experiment.

Chemicals:

All the chemicals were of highest purity and were purchased from Sigma Chemical Company, USA, BDH Chemical Pool, England or from E. Merck, Darmstadt. Monosodium glutamate (99%) purchased from Ajinomoto Co. Inc., Tokyo, Japan, dissolved in 0.9% NaCl was injected intraperitoneally at doses of 4 g/kg body weight to test animals. Control animals received 0.9% NaCl. The animals were decapitated 2 hours later and samples of plasma were collected. Fresh liver samples were immediately used.

Estimation of tryptophan pyrrolase activity:

Animals decapitated and fresh livers removed immediately were purged of blood by perfusion with ice cold saline. Tryptophan pyrrolase activity in fresh liver homogenate was determined by the method of Feigelson and Greengard (1961) either in the absence (holoenzyme activity) or in presence (total enzyme activity) of added 2 (μ m) haematin as described by Haleem et al. (1984) and Haleem (1990). The apoenzyme activity was obtained by the difference.

Determination of total and free tryptophan in plasma:

Plasma total and free (ultrafiltrable) tryptophan was determined by the fluorimetric method of Denckla & Dewey (1967) as modified by Bloxam & Warren (1974).

Determination of plasma corticosterone levels:

The fluorimetric method of Mattingley (1962) as described by Haleem (1988) was used for the determination of corticosterone in plasma.

Statistical analysis:

The data on the effect of MSG on tryptophan pyrrolase activity, plasma tryptophan

tophan and plasma corticosterone levels were analysed by two-way ANOVA (factor 1 sex, factor 2 drug and interaction between two factors). Individual comparisons were made using Newman-Keuls statistics. Differences between groups were considered significant when $P < 0.05$.

RESULTS

Table I shows hepatic tryptophan pyrrolase activity in male and female rats 2 hours after the intraperitoneal administration of 4 g/kg MSG. Data analysed by two-way ANOVA show significant effect of sex, drug and sex x drug interaction on holoenzyme activity ($P < 0.01$, $P < 0.01$ and $P < 0.05$) respectively. Drug effect was also significant for apoenzyme and total enzyme.

Individual comparison by Newman-Keuls statistics shows significant increase (61.6% $P < 0.05$) in holoenzyme activity in MSG treated male rats. In female rats the increase (24.2%) was not significant. Apoenzyme activity was decreased comparably (63.6% $P < 0.01$) in the two sexes. Total enzyme activity was significantly decreased (37.59%, $P < 0.01$) only in female rats. It was also decreased (20.6%) in male rats but the differences were non-significant.

Table II shows the effect of MSG on plasma total and free tryptophan concentration and plasma corticosterone levels in male and female rats. Data analysed by two-way ANOVA shows significant effect of sex ($F = 9.64$, $P < 0.01$) and drug ($F = 21.29$, $P < 0.01$) on plasma total tryptophan levels. Sex x drug interaction was not significant. For plasma free tryptophan concentration only the effect of drug was significant ($F = 49.93$, $P < 0.01$). Individual comparison by Newman-Keuls showed an increase (51.2%, $P < 0.01$) in plasma total tryptophan levels in female rats. These increases were non-significant in male rats. Free tryptophan concentrations were decreased significantly ($P < 0.01$) in both sexes.

Fig. 1 shows effect of MSG on plasma corticosterone levels. Individual comparison by Newman-Keuls shows that there was significant increase 98.1%; $P < 0.01$) in plasma corticosterone levels in MSG treated male rats while in female there was 23.4% increase which was not significant.

DISCUSSION

The increase in holotryptophan pyrrolase activity after MSG treatment in male rats observed in the present study may be due to elevated plasma corticosterone levels after the treatment. It has been shown that MSG administration (4 g/kg) i.p. increases plasma levels of corticosterone in male rats (Dolnicoff, 1988; Magarinos, 1988). Our

data shows that 2 hours after the administration the increases were significant in male but not in female rats. The sex difference in the effects of MSG on plasma corticosterone level (Fig. 1) may be due to the known effect of drug on serum estrogen levels. MSG treatment has been shown to decrease serum estrogen levels (Chin Hsu et al., 1990) in female rats. A decrease in apotryptophan pyrrolase activity after MSG treatment may be caused by either an inhibition of apoenzyme synthesis or inactivation of the existing apoenzyme by prevention of its conjugation with haem or by an allosteric mechanism (Cho-Chung & Pitot, 1967).

The present study shows that administration of MSG increases holoenzyme and decreases apoenzyme activity in male rats. Thus total enzyme activity is not affected. A decrease in total tryptophan pyrrolase activity by MSG in females may be due to the inability of MSG to increase plasma level of corticosterone in females. Factors which inhibit liver tryptophan pyrrolase activity have been shown to increase plasma tryptophan concentration (Badawy & Evans, 1991). The significant increase in total plasma tryptophan in females may also be due to decreased utilization by liver. Thus due to non-significant inhibition of total tryptophan pyrrolase activity the increase of plasma total tryptophan was also not significant in MSG treated male rats.

Factors which increases plasma free fatty acid have been shown to increase free tryptophan in plasma (Curzon & Knott, 1974). The decrease of plasma free tryptophan in the present study may also be due to decreased level of free fatty acid.

The reduction in growth hormone after MSG treatment (Pizzi et al., 1977; Maiater et al., 1991) may account in part for reduction in plasma non-esterified fatty acid levels, therefore plasma free tryptophan concentration was decreased in both sexes. It may be possible that MSG treatment might have reduced plasma non-esterified fatty acid levels thus promoting tryptophan binding to albumin leading to elevation in the size of plasma bound tryptophan. Because changes in tryptophan pyrrolase activity and plasma tryptophan levels are often linked with the synthesis of brain serotonin (Badawy, 1977) and MSG administration alters the synthesis of 5-hydroxytryptamine (5-HT) in brain (Johnston et al., 1984). The present study tends to suggest that MSG administration may have different effects on the biosynthesis of 5-HT in male and female sex.

Table I
Effect of monosodium glutamate (MSG) administration (4 g/kg)
i.p. on hepatic tryptophan pyrrolase activity
(ug of kynurenine produced/h/g wet wt.)
in male and female rats

Determination	Male		Female		Two-Way ANOVA DF (1,12)		
	Saline	MSG	Saline	MSG	Sex	Drug	Sex x Drug
Total enzyme	5.00 ± 0.28	3.97 ± 0.65	5.32 ± 0.68	3.32 ± 0.22*	F = 0.41	F = 35.80	F = 3.72
		- 20.60%		- 37.59%	N.S.	P < 0.01	N.S.
Holoenzyme	1.67 ± 0.09	2.70 ± 0.34**	1.57 ± 0.22	1.95 ± 0.09 ⁺	F = 15.34	F = 41.62	F = 8.97
		+ 61.60%		+ 24.20%	P < 0.01	P < 0.01	P < 0.05
Apoenzyme	3.32 ± 0.29	1.20 ± 0.16**	3.77 ± 0.55	1.37 ± 0.17**	F = 3.44	F = 180.33	F = 0.66
		- 63.60%		- 63.60%	N.S.	P < 0.01	N.S.

Values are means ± S.D. ** P < 0.01 from respective controls, + P < 0.01 from similarly treated male rats by Newman-Keuls Q statistics following two-way ANOVA.

Table II
Effect of monosodium glutamate (MSG) administration (4 g/kg) i.p. on plasma
tryptophan concentration (ug/ml) in male, 2 hours after injection

Determination	Male		Female		Two-Way ANOVA DF (1,20)		
	Saline	MSG	Saline	MSG	Sex	Drug	Sex x Drug
Total	20.35 ± 3.69	27.71 ± 6.31	24.40 ± 3.25	**36.90 ± 5.30 ⁺	F=9.64	F=21.29	F=1.40
		+ 36.10%	+	51.20%	P<0.01	P<0.01	N.S.
Free	1.51 ± 0.17 ^{**}	1.01 ± 0.17 ^{**}	1.46 ± 0.16	**1.03 ± 0.13	F=0.063	F=49.93	F=0.25
		- 33.10%		- 29.40%	N.S.	P<0.01	N.S.
% Free	7.59 ± 1.06	**3.83 ± 1.03	6.03 ± 0.54 ⁺	**2.85 ± 0.58	F=13.66	F=101.78	F=0.70
		- 49.50%	-	52.70%	P<0.01	P<0.01	N.S.

Values are means ± S.D. (n=6) ****P** < 0.01 from respective controls, + P < 0.01 from similarly treated male rats by Newman-Keuls Q statistics following two-way ANOVA.

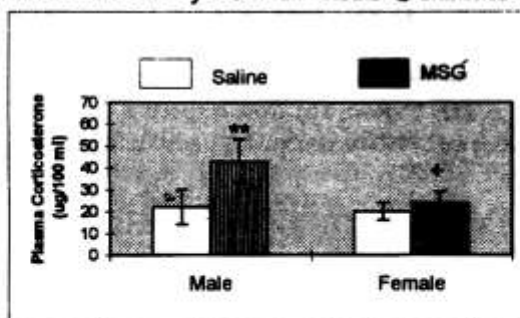


Fig. 1 shows the effect of plasma corticosterone levels in rats 2 hrs. after injection. Values are mean ± S.D. (n=6) ****P** 0.01 from respective controls, + P 0.01 from similarly treated male rats by Newman-Kuels Q Statistics following two way ANOVA.

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